

Question 1

Question-1:

Create 1 Public Docker Hub registry named flask_webapp_yourname.

- Clone below repository on your system.
<https://github.com/mmumshad/simple-webapp-docker.git>
- Initialize a local repository & copy the code from above repo to your local repository in your working branch.
- Once code is copied to the local repository, build the docker image & add meaningful tags with version 1 and push to Docker Hub registry.
- Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , labels app: flask-webapp , containerPort: 8080 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select flask-webapp with port 8080 & targetPort 8080 with any nodePort between range 30000-32768.
- Once a service is created try accessing the web page in the browser as below. (30011 is nodeport mentioned in service.yaml). Meanwhile open app.py from your code to understand paths & output.
http://master_ip:30011/
http://master_ip:30011/how are you
- Now , update the app.py from your code and add below route above if **name == "main"** line
@app.route('/Who are you')
def cloudeithix():
return 'Yes, I am cloudeithix, and You !!!'
- Once the file is updated , rebuild the docker image & add meaningful tags with version 2 and push to Docker Hub registry.
- Now we have the latest docker image in repo, It's time to roll out a new image. Roll out the new Image with all three ways one by one.
 1. With kubectl set command
 2. With kubectl edit deployment
 3. With deployment.yaml file modification.
- Run the # kubectl rollout command to check status and history.
- Note:- Once above step 1 is done , run # kubectl rollout undo deployment command to rollback the change and then try a second way of rollout.
- In the browser run all three routes & notice the changes.
http://master_ip:30011/
http://master_ip:30011/how are you
http://master_ip:30011/Who are you
- Once done with all above steps , commit all the changes to the remote repository.
- **Capture the snap and prepare a well formatted document.**

System / Terminal Output

```
payal_kharat@LAPTOP-# git clone https://github.com/mmumshad/simple-webapp-docker.git
Cloning into 'simple-webapp-docker'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 14 (delta 3), reused 2 (delta 2), pack-reused 7 (from 1)
Receiving objects: 100% (14/14), done.
Resolving deltas: 100% (3/3), done.
```

```
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
hint: warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/.git/
```

```
payal_kharat@LAPTOP-# cp -rf simple-webapp-docker/* .
payal_kharat@LAPTOP-# ls
Dockerfile app.py simple-webapp-docker
payal_kharat@LAPTOP-# podman build -t flask_webapp_payal:v1 .
STEP 1/5: FROM ubuntu:20.04
Resolved "ubuntu" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/ubuntu:20.04...
Getting image source signatures
Copying blob 13b7e930469f done    |
Copying config b7bab04fd9 done    |
Writing manifest to image destination
STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
```

```
payal_kharat@LAPTOP-# podman tag localhost/flask_webapp_payal:v1
docker.io/payalkharat/flask_webapp_payal:v1
payal_kharat@LAPTOP-# podman push docker.io/payalkharat/flask_webapp_payal:v1
Getting image source signatures
Copying blob 136df475df75 done    |
Copying blob b221cfe8484e done    |
Copying blob aed4789f5432 done    |
Copying blob 13b7e930469f skipped: already exists
Copying config 29aa696b24 done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
```

```
metadata:
  name: flask-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp
  template:
    metadata:
      labels:
        app: flask-webapp
    spec:
      containers:
        - name: flask-webapp
          image: docker.io/payalkharat/flask_webapp_payal:v1
          ports:
            - containerPort: 8080
```

```
-----
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/flask-webapp created
payal_kharat@LAPTOP-# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
flask-webapp        3/3     3             3           3m19s
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-webapp-7d5fbbd947-678d9      1/1     Running   0           55s
flask-webapp-7d5fbbd947-f22mr      1/1     Running   0           55s
flask-webapp-7d5fbbd947-kn8mk      1/1     Running   0           55s
-----
```

```
payal_kharat@LAPTOP-# cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: flask-webapp-service
spec:
  type: NodePort
  selector:
    app: flask-webapp
  ports:
    - port: 8080
      targetPort: 8080
      nodePort: 30011
-----
```

```
payal_kharat@LAPTOP-# kubectl apply -f service.yaml
service/flask-webapp-service created
payal_kharat@LAPTOP-# kubectl get svc
NAME                TYPE           CLUSTER-IP      EXTERNAL-IP  PORT(S)
AGE
cloudethix-clusterip ClusterIP       10.96.172.109    <none>       80/TCP
4h12m
flask-webapp-service NodePort        10.96.219.54     <none>       8080:30011/TCP
38s
-----
```

```
payal_kharat@LAPTOP-# cat app.py
import os
from flask import Flask
app = Flask(__name__)

@app.route("/")
```

```
def main():
    return "Welcome!"

@app.route('/how are you')
def hello():
    return 'I am good, how about you?'

@app.route('/Who are you')
def cloudehix():
    return 'Yes, I am cloudehix, and You !!!'

if __name__ == "__main__":
    app.run()
```

```
payal_kharat@LAPTOP-# podman build -t flask_webapp_payal:v2 .
STEP 1/5: FROM ubuntu:20.04
STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
--> Using cache
3284c28d44b5a4c83f437142a15eb81b21e2d446de186b3e8615cfa34a25d54a
--> 3284c28d44b5
STEP 3/5: RUN pip install flask
--> Using cache
4dc49a101f7ed306dab744218c77bf01fc4e03656c121ddfae365582c038078d
--> 4dc49a101f7e
STEP 4/5: COPY app.py /opt/
--> f2b14aae8f8e
STEP 5/5: ENTRYPOINT FLASK_APP=/opt/app.py flask run --host=0.0.0.0 --
port=8080
COMMIT flask_webapp_payal:v2
--> 2eb5d0f90730
Successfully tagged localhost/flask_webapp_payal:v2
2eb5d0f907305961efa22039da30b998d85bdf8f33e298baf581deb347a64bfd
```

```
payal_kharat@LAPTOP-# podman tag flask_webapp_payal:v2
docker.io/payalkharat/flask_webapp_payal:v2
payal_kharat@LAPTOP-# podman push docker.io/payalkharat/flask_webapp_payal:v2
Getting image source signatures
Copying blob c18f10e5e7ec done    |
Copying blob e7ceb2b17c6a skipped: already exists
Copying blob 13b7e930469f skipped: already exists
Copying blob 454ea61b84a9 skipped: already exists
Copying config 2eb5d0f907 done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
flask-webapp-7d5fbbd947-678d9	1/1	Running	0	34m
flask-webapp-7d5fbbd947-f22mr	1/1	Running	0	34m
flask-webapp-7d5fbbd947-kn8mk	1/1	Running	0	34m

```
payal_kharat@LAPTOP-# kubectl get deployment
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
flask-webapp	3/3	3	3	34m

```
payal_kharat@LAPTOP-# kubectl set image deployment/flask-webapp flask-
webapp=docker.io/payalkharat/flask_webapp_payal:v2
deployment.apps/flask-webapp image updated
```

```
payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
```

```
-----
payal_kharat@LAPTOP-# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
flask-webapp        3/3     3             3            35m
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS      RESTARTS   AGE
flask-webapp-6687cd68c6-hnz8t      1/1     Running     0           34s
flask-webapp-6687cd68c6-jkmqx      1/1     Running     0           29s
flask-webapp-6687cd68c6-lsbvh      1/1     Running     0           41s
flask-webapp-7d5fbbd947-678d9      1/1     Terminating 0           35m
flask-webapp-7d5fbbd947-f22mr      1/1     Terminating 0           35m
-----

payal_kharat@LAPTOP-# kubectl rollout history deployment/flask-webapp
deployment.apps/flask-webapp
REVISION  CHANGE-CAUSE
1          <none>
2          <none>
payal_kharat@LAPTOP-# kubectl rollout undo deployment/flask-webapp
deployment.apps/flask-webapp rolled back
payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
-----

payal_kharat@LAPTOP-# vim deployment.yaml
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp
  template:
    metadata:
      labels:
        app: flask-webapp
    spec:
      containers:
        - name: flask-webapp
          image: docker.io/payalkharat/flask_webapp_payal:v2
          ports:
            - containerPort: 8080
-----

payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/flask-webapp configured
payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
-----

payal_kharat@LAPTOP-# kubectl rollout history deployment/flask-webapp
deployment.apps/flask-webapp
REVISION  CHANGE-CAUSE
3          <none>
4          <none>
-----

payal_kharat@LAPTOP-# kubectl rollout undo deployment/flask-webapp
deployment.apps/flask-webapp rolled back
-----

payal_kharat@LAPTOP-# kubectl edit deployment flask-webapp
deployment.apps/flask-webapp edited
-----
```

```
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-webapp-6687cd68c6-64k2x      1/1     Running   0           18s
flask-webapp-6687cd68c6-pcvss      1/1     Running   0           16s
flask-webapp-6687cd68c6-sdx6t      1/1     Running   0           15s
flask-webapp-7d5fbbd947-gr7dj      1/1     Terminating   0           2m13s
flask-webapp-7d5fbbd947-k66gz      1/1     Terminating   0           2m11s
flask-webapp-7d5fbbd947-vcsj6      1/1     Terminating   0           2m11s
-----
payal_kharat@LAPTOP-# kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
payal_kharat@LAPTOP-# kubectl rollout history deployment/flask-webapp
deployment.apps/flask-webapp
REVISION   CHANGE-CAUSE
5          <none>
6          <none>
```

Question 3

Question-3:

Create 1 Public Docker Hub registry named cloudethix_configmap_yourname.

- Clone below repository on your system.
<https://github.com/zembutsu/docker-sample-nginx.git>
- Initialize a local repository & copy the code from above repo to your local repository in the working branch.
- Once code is copied , build a docker image from docker file and add meaningful tags and push to docker hub repository.
- Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , labels app: frontend-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select frontend-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Once the service is created try accessing the web page in the browser as below. Notice the changes & take the snap.
- Now create a configmap.yaml file with below data & delete the deployment that you have created.

I am Cloudethix Team, Are you ?!!

Version: 1.1

- Then update the same deployment.yaml file and mount configmap as volume on container using volumeMounts with mountPath /usr/share/nginx/html/
- Now it's time to create configmap & deployment. Once created , try to access the webpage in the browser & confirm that the index page is the same as we have in configmap.

=====

System / Terminal Output

```
payal_kharat@LAPTOP-# git clone https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
```

```
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
```

```
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-3/.git/
```

```
payal_kharat@LAPTOP-# cp -rf docker-sample-nginx/* .
```

```
payal_kharat@LAPTOP-# podman build -t cloudethix_configmap_payal:1.0 .
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
bfba633494f51cc913622ffd1987833f35e1c8cff0ccf02885633eeca04dc5a0
--> bfba633494f5
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT cloudethix_configmap_payal:1.0
--> 71c2e8cca826
Successfully tagged localhost/cloudethix_configmap_payal:1.0
71c2e8cca8262f19f3288b58a6adb3b921b873e24219b011a4bb9a1f2c1dfdce
```

```
payal_kharat@LAPTOP-# podman tag cloudethix_configmap_payal:1.0
docker.io/payalkharat/cloudethix_configmap_payal:1.0
payal_kharat@LAPTOP-# podman push
docker.io/payalkharat/cloudethix_configmap_payal:1.0
Getting image source signatures
Copying blob d94291c26261 skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob 0d62d88506ba skipped: already exists
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob 8cdfef4f23778 skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob e43474696a32 done    |
Copying blob da4dea1b00af skipped: already exists
Copying blob fb597529c916 skipped: already exists
Copying blob b83fc70afc5e skipped: already exists
Copying config 71c2e8cca8 done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat deployment.yaml service.yaml
apiVersion: apps/v1
```

```
kind: Deployment
metadata:
  name: frontend-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp
  template:
    metadata:
      labels:
        app: frontend-webapp
    spec:
      containers:
        - name: frontend-webapp
          image: docker.io/payalkharat/cloudethix_configmap_payal:1.0
          ports:
            - containerPort: 80
apiVersion: v1
kind: Service
metadata:
  name: frontend-webapp-service
spec:
  type: NodePort
  selector:
    app: frontend-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30009
```

```
-----
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/frontend-webapp created
payal_kharat@LAPTOP-# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
frontend-webapp     3/3     3             3           3m57s
```

```
-----
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
frontend-webapp-5667d88db8-bxfwr   1/1     Running   0           6m8s
frontend-webapp-5667d88db8-sl5zf    1/1     Running   0           6m8s
frontend-webapp-5667d88db8-slzdl    1/1     Running   0           6m8s
```

```
-----
payal_kharat@LAPTOP-# kubectl apply -f service.yaml
service/frontend-webapp-service created
payal_kharat@LAPTOP-# kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)
AGE
frontend            NodePort    10.96.77.238    <none>
80:30008/TCP        4h52m
frontend-webapp-service NodePort    10.96.201.214   <none>
80:30009/TCP        97s
```

```
-----
payal_kharat@LAPTOP-# cat configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: frontend-config
data:
```

```
index.html: |
<html>
<body>
<h1> I am Cloudethix Team, Are you ?!! </h1>
Version: 1.1
</body>
</html>
```

```
payal_kharat@LAPTOP-# kubectl apply -f configmap.yaml
configmap/frontend-config created
payal_kharat@LAPTOP-# kubectl get configmap
NAME          DATA   AGE
frontend-config    1      6s
kube-root-ca.crt  1      6h41m
```

```
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp
  template:
    metadata:
      labels:
        app: frontend-webapp
    spec:
      containers:
        - name: frontend-webapp
          image: docker.io/payalkharat/cloudethix_configmap_payal:1.0
          ports:
            - containerPort: 80
          volumeMounts:
            - name: frontend-html
              mountPath: /usr/share/nginx/html/
      volumes:
        - name: frontend-html
          configMap:
            name: frontend-config
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/frontend-webapp created
payal_kharat@LAPTOP-# kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
frontend-webapp  3/3     3            3           6s
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
frontend-webapp-65c5f6c4fd-cbq65    1/1     Running   0           11s
frontend-webapp-65c5f6c4fd-grcld    1/1     Running   0           10s
frontend-webapp-65c5f6c4fd-qqp7r    1/1     Running   0           10s
```

Question 4

Question-4:

- Create 1 Public Docker Hub registry named cloudethix_multicontainer_yourname.

- Clone below repository on your system.
<https://github.com/janakiramm/Kubernetes-multi-container-pod.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branches.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
- Now go to the deploy directory and notice the files.
- Here, web-pod-1.yml file will create the pod with two containers (Multi container). Take a note of lables , name of containers and ports. Also, please make sure you will update the python container image that you have pushed to your docker registry.
- Now, open web-svc.yml file and notice service Type , selectors & targetPort. Apply the file.
- Now open db-pod.yml & notice the lables , name , Image, containerPort and apply the file.
- Now open the db-svc.yml file and notice service Type , selectors & targetPort. Apply the file.
- Once above files are applied , Check that the Pods and Services are created using command line or lens.
- Now , from the command line run below urls & notice the changes.

curl http://*NODE_IP* :NODE_PORT**/init**

Initialize the database with sample schema

- Now it's time to Insert some sample data. Make sure you will use correct *NODE_IP* :*NODE_PORT*

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "1",
"user": "John Doe"}' http://*NODE_IP* :*NODE_PORT*/users/add**

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "2",
"user": "Jane Doe"}' http://*NODE_IP* :*NODE_PORT*/users/add**

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "3",
"user": "Bill Colls"}' http://*NODE_IP* :*NODE_PORT*/users/add**

**curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "4",
"user": "Mike Taylor"}' http://*NODE_IP* :*NODE_PORT*/users/add**

- Now access the data that we have added to database using below command.

curl http://*NODE_IP* :NODE_PORT**/users/1**

- The second time you access the data, it appends '(c)' indicating that it is pulled from the Redis cache.
- Also, try to access mysql shell i.e db pod & run select * from the users table. check app.py for DB related information.
- Prepare proper documentation in brief & write start to end flow. Refer below link if you face any issues.

<https://github.com/janakiramm/Kubernetes-multi-container-pod>

=====

System / Terminal Output

```
payal_kharat@LAPTOP-# git clone https://github.com/janakiramm/Kubernetes-
multi-container-pod.git
Cloning into 'Kubernetes-multi-container-pod'...
remote: Enumerating objects: 51, done.
remote: Total 51 (delta 0), reused 0 (delta 0), pack-reused 51 (from 1)
Receiving objects: 100% (51/51), 88.14 KiB | 876.00 KiB/s, done.
Resolving deltas: 100% (21/21), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-4$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-4/.git/
-----
payal_kharat@LAPTOP-# cp -rf Kubernetes-multi-container-pod/* .
payal_kharat@LAPTOP-# ls
Build  Deploy  Kubernetes-multi-container-pod  README.md  multi-container-
pod.png
payal_kharat@LAPTOP-# git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Build/
    Deploy/
    Kubernetes-multi-container-pod/
    README.md
    multi-container-pod.png
nothing added to commit but untracked files present (use "git add" to track)
-----
payal_kharat@LAPTOP-# git add .
warning: adding embedded git repository: Kubernetes-multi-container-pod
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> Kubernetes-multi-container-pod
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached Kubernetes-multi-container-pod
hint:
```

hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"

```
payal_kharat@LAPTOP-# git commit -m "add Kubernetes multi-container
application"
[master (root-commit) b90b4c4] add Kubernetes multi-container  application
13 files changed, 280 insertions(+)
create mode 100644 Build/Dockerfile
create mode 100644 Build/app.py
create mode 100644 Build/docker-compose.yml
create mode 100644 Build/requirements.txt
create mode 100644 Deploy/db-pod.yml
create mode 100644 Deploy/db-svc.yml
create mode 100644 Deploy/web-pod-1.yml
create mode 100644 Deploy/web-pod-2.yml
create mode 100644 Deploy/web-rc.yml
create mode 100644 Deploy/web-svc.yml
create mode 160000 Kubernetes-multi-container-pod
create mode 100644 README.md
create mode 100644 multi-container-pod.png
```

```
payal_kharat@LAPTOP-# podman build -t cloudethix-multicontainer-python:v1 .
STEP 1/3: FROM python:2.7-onbuild
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:2.7-onbuild...
Getting image source signatures
Copying blob 46dde23c37b3 done    |
Copying blob e7428f935583 done    |
Copying blob 6ebaeb074589 done    |
Copying blob 0c3de61682aa done    |
Copying blob d660b1f15b9b done    |
Copying blob 56f10ddf1173 done    |
Copying blob 4473537c621d done    |
Copying blob 3106f7df3d1c done    |
Copying blob 3de1c6ceef68 done    |
Copying config 3f246dd60a done    |
```

```
payal_kharat@LAPTOP-# podman tag cloudethix-multicontainer-python:v1
payalkharat/cloudethix_multicontainer_payal:python-v1
payal_kharat@LAPTOP-# podman push
payalkharat/cloudethix_multicontainer_payal:python-v1
Getting image source signatures
Copying blob d660b1f15b9b skipped: already exists
Copying blob 56f10ddf1173 skipped: already exists
Copying blob e7428f935583 skipped: already exists
Copying blob 46dde23c37b3 skipped: already exists
Copying blob 0c3de61682aa skipped: already exists
Copying blob 6ebaeb074589 skipped: already exists
Copying blob 3de1c6ceef68 skipped: already exists
Copying blob 6ba9395573b3 done    |
Copying blob 92c3866eed4c skipped: already exists
Copying blob 8b5ad887043b skipped: already exists
Copying blob 3106f7df3d1c skipped: already exists
Copying blob 4473537c621d skipped: already exists
Copying config 0836e0ab53 done    |
Writing manifest to image destination
```

```
payal_kharat@LAPTOP-# cat web-pod-1.yml
apiVersion: "v1"
```

```
kind: Pod
metadata:
  name: web1
  labels:
    name: web
    app: demo
spec:
  containers:
    - name: redis
      image: redis
      ports:
        - containerPort: 6379
          name: redis
          protocol: TCP
    - name: python
      image: payalkharat/cloudethix_multicontainer_payal:python-v1
      env:
        - name: "REDIS_HOST"
          value: "localhost"
      ports:
        - containerPort: 5000
          name: http
          protocol: TCP
```

```
-----
payal_kharat@LAPTOP-# cat web-svc.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: web
  labels:
    name: web
    app: demo
spec:
  selector:
    name: web
  type: NodePort
  ports:
    - port: 80
      name: http
      targetPort: 5000
      protocol: TCP
```

```
-----
payal_kharat@LAPTOP-# kubectl apply -f web-svc.yml
```

```
service/web created
```

```
payal_kharat@LAPTOP-# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	12h
web	NodePort	10.96.196.49	<none>	80:31143/TCP	6s

```
payal_kharat@LAPTOP-# cat db-svc.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
spec:
  ports:
    - port: 3306
      name: mysql
```

```
targetPort: 3306
selector:
  name: mysql
  app: demo
-----
payal_kharat@LAPTOP-# kubectl apply -f db-svc.yml
service/mysql created
payal_kharat@LAPTOP-# kubectl get svc
NAME            TYPE            CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes      ClusterIP        10.96.0.1        <none>           443/TCP          12h
mysql           ClusterIP        10.96.91.36      <none>           3306/TCP         5s
payal_kharat@LAPTOP-# cat db-pod.yml
apiVersion: "v1"
kind: Pod
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
spec:
  containers:
    - name: mysql
      image: mysql:5.7.25
      ports:
        - containerPort: 3306
          protocol: TCP
      env:
        -
          name: "MYSQL_ROOT_PASSWORD"
          value: "password"
-----
payal_kharat@LAPTOP-# kubectl apply -f db-pod.yml
pod/mysql created
payal_kharat@LAPTOP-# kubectl get pods
NAME    READY   STATUS             RESTARTS   AGE
mysql   1/1     Running            0          19s
payal_kharat@LAPTOP-# kubectl get pods
NAME    READY   STATUS             RESTARTS   AGE
mysql   1/1     Running            0          88s
web1    2/2     Running            1 (36s ago) 102s
payal_kharat@LAPTOP-# kubectl get svc
NAME            TYPE            CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes      ClusterIP        10.96.0.1        <none>           443/TCP          4m52s
mysql           ClusterIP        10.96.233.135    <none>           3306/TCP         2m34s
web            NodePort         10.96.48.40      <none>           80:32026/TCP     2m48s
-----
payal_kharat@LAPTOP-# curl http://localhost:8080/init
DB Init done
payal_kharat@LAPTOP-# curl -i -H "Content-Type: application/json" -X POST -d
'{"uid": "1", "user": "John Doe"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:07:43 GMT

Added
-----
payal_kharat@LAPTOP-# curl -i \> curl -i \
-H "Content-Type: application/json" \
```

```
-X POST \
-d '{"uid":"2","user":"Jane Doe"}' \
http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:11:45 GMT
```

```
payal_kharat@LAPTOP-# curl -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"3","user":"Bill Colls"}' \
http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:11:45 GMT
```

```
payal_kharat@LAPTOP-# curl -i \ -i \
-H "Content-Type: application/json" \
-X POST \
-d '{"uid":"4","user":"Mike Taylor"}' \
http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:12:37 GMT
```

```
payal_kharat@LAPTOP-# loy$ http://localhost:8080/users/1
John Doe
payal_kharat@LAPTOP-# curl http://localhost:8080/users/1
John Doe(c)
payal_kharat@LAPTOP-# curl http://localhost:8080/users/2
Jane Doe
payal_kharat@LAPTOP-# curl http://localhost:8080/users/3
Bill Colls
```

```
payal_kharat@LAPTOP-# curl http://localhost:8080/users/3
Mike Taylor
payal_kharat@LAPTOP-# kubectl exec -it mysql -- bash
root@mysql:/# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 4
Server version: 5.7.25 MySQL Community Server (GPL)
```

Copyright (c) 2000, 2019, Oracle and/or its affiliates. All rights reserved.

Oracle is a registered trademark of Oracle Corporation and/or its affiliates. Other names may be trademarks of their respective owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> SHOW DATABASES;
+-----+
| Database          |
```

```
+-----+
| information_schema |
| USERDB             |
| mysql              |
| performance_schema |
| sys                |
+-----+
5 rows in set (0.01 sec)
mysql> SHOW DATABASES;
+-----+
| Database           |
+-----+
| information_schema |
| USERDB             |
| mysql              |
| performance_schema |
| sys                |
+-----+
5 rows in set (0.01 sec)

-----

mysql> USE USERDB;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_USERDB   |
+-----+
| users              |
+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM users;
+-----+-----+
| ID  | USER          |
+-----+-----+
| 1   | John Doe      |
| 2   | Jane Doe      |
| 3   | Bill Colls    |
| 4   | Mike Taylor   |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

Question 5

Question-5:

Create 1 Public Docker Hub registry named cloudethix_Initcontainer_yourname.

- Clone below repository on your system.
<https://github.com/janakiramm/simpleapp.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.

- Once Images are pushed to Docker hub registries, create a directory named kube. Inside the kube directory create deployment.yaml file with 3 replication , label app: simpleapp-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.
- Create one service.yaml file with type nodeport & select simpleapp-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Open the webpage in the browser and notice the changes and capture the snap.
- Then delete the deployment that you have just created.
- Update the deployment.yaml file and add volumeMounts with mountPath /usr/share/nginx/html from emptyDir: {} volume.
- Once above changes are added, add initContainers block with below parameters. Also add volumeMounts for Init Container with mountPath "/work-dir" from emptyDir: {} volume.
initContainers:
 - name: install
 - image: busybox:1.28
 - command:
 - wget
 - "-O"
 - "/work-dir/index.html"
 - <http://info.cern.ch>
 - volumeMounts:
 - name: workdir
 - mountPath: "/work-dir"
- Add volumes with emptyDir: {} in deployment.yaml file.
- Once the deployment.yaml file is ready, create the deployment & access the page in the browser and notice the changes.

=====

System / Terminal Output

```
payal_kharat@LAPTOP-# git clone https://github.com/janakiramm/simpleapp.git
Cloning into 'simpleapp'...
remote: Enumerating objects: 47, done.
remote: Total 47 (delta 0), reused 0 (delta 0), pack-reused 47 (from 1)
Receiving objects: 100% (47/47), 8.20 KiB | 335.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-5$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
```

```
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-5/.git/

-----

payal_kharat@LAPTOP-# cp -rf simpleapp/* .
payal_kharat@LAPTOP-# ls
Dockerfile  html  simpleapp  wrapper.sh

-----

payal_kharat@LAPTOP-# podman build -t cloudethix-initcontainer:simpleapp-v1 .
STEP 1/4: FROM docker.io/library/nginx
Trying to pull docker.io/library/nginx:latest...
Getting image source signatures
Copying blob 02fc02c4ab8d done    |
Copying blob 6310eb16bf42 done    |
Copying blob c12f394dea35 done    |
Copying blob a44b5c8be616 done    |
Copying blob 956faab5efb3 done    |
Copying blob 07db7bf2649b done    |
Copying blob f340c1b7c1d6 done    |
Copying config 5f52b9aca7 done    |
Writing manifest to image destination
STEP 2/4: COPY wrapper.sh /
--> f87d805d1caa
STEP 3/4: COPY html /usr/share/nginx/html
--> e9bb2a7f6bce
STEP 4/4: CMD ["/wrapper.sh"]
COMMIT cloudethix-initcontainer:simpleapp-v1
--> 8c761ea81c19
Successfully tagged localhost/cloudethix-initcontainer:simpleapp-v1
8c761ea81c19787fc1a5e92ecd523314ce8eedc0617127528bfa465386622712

-----

payal_kharat@LAPTOP-# podman tag cloudethix-initcontainer:simpleapp-v1
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
payal_kharat@LAPTOP-# podman login docker.io
Username: payalkharat
Password:
Login Succeeded!

-----

payal_kharat@LAPTOP-# podman push
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
Getting image source signatures
Copying blob 07db7bf2649b skipped: already exists
Copying blob 02fc02c4ab8d skipped: already exists
Copying blob a44b5c8be616 skipped: already exists
Copying blob 956faab5efb3 skipped: already exists
Copying blob 6310eb16bf42 skipped: already exists
Copying blob e4a3debe156f done    |
Copying blob 966b8902a23a done    |
Copying blob c12f394dea35 skipped: already exists
Copying blob f340c1b7c1d6 skipped: already exists
Copying config 8c761ea81c done    |
Writing manifest to image destination

-----

payal_kharat@LAPTOP-# cat deployment.yaml service.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp
spec:
  replicas: 3
```

```
selector:
  matchLabels:
    app: simpleapp-webapp
template:
  metadata:
    labels:
      app: simpleapp-webapp
  spec:
    containers:
      - name: simpleapp
        image:
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
        ports:
          - containerPort: 80
apiVersion: v1
kind: Service
metadata:
  name: simpleapp-webapp-service
spec:
  type: NodePort
  selector:
    app: simpleapp-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
```

```
-----
payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml
deployment.apps/simpleapp-webapp created
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
simpleapp-webapp-6564955fc-frj4r    1/1     Running   0           24s
simpleapp-webapp-6564955fc-g65dk    1/1     Running   0           24s
simpleapp-webapp-6564955fc-w8r9b    1/1     Running   0           24s
-----
```

```
payal_kharat@LAPTOP-# kubectl apply -f service.yaml
service/simpleapp-webapp-service created
payal_kharat@LAPTOP-# kubectl get svc
NAME                                TYPE           CLUSTER-IP    EXTERNAL-IP  PORT(S)
AGE
kubernetes                          ClusterIP      10.96.0.1     <none>       443/TCP
48m
simpleapp-webapp-service             NodePort       10.96.232.18  <none>       80:30080/TCP   30s
-----
```

```
payal_kharat@LAPTOP-# kubectl delete deployment simpleapp-webapp
deployment.apps "simpleapp-webapp" deleted from default namespace
payal_kharat@LAPTOP-# kubectl get pods
No resources found in default namespace.
payal_kharat@LAPTOP-# cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp
  template:
```

```

metadata:
  labels:
    app: simpleapp-webapp
spec:

  initContainers:
    - name: install
      image: busybox:1.28
      command:
        - wget
        - "-O"
        - "/work-dir/index.html"
        - http://info.cern.ch
      volumeMounts:
        - name: workdir
          mountPath: "/work-dir"

  containers:
    - name: simpleapp
      image:
docker.io/payalkharat/cloudethix_initcontainer_payal:simpleapp-v1
      ports:
        - containerPort: 80
      volumeMounts:
        - name: workdir
          mountPath: /usr/share/nginx/html

  volumes:
    - name: workdir
      emptyDir: {}

```

```

payal_kharat@LAPTOP-# kubectl apply -f deployment.yaml

```

```

deployment.apps/simpleapp-webapp created

```

```

payal_kharat@LAPTOP-# kubectl get deployment

```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
simpleapp-webapp	3/3	3	3	12s

```

payal_kharat@LAPTOP-# kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
simpleapp-webapp-7748bd6596-2tp92	1/1	Running	0	17s
simpleapp-webapp-7748bd6596-f6dv2	1/1	Running	0	17s
simpleapp-webapp-7748bd6596-zw5hn	1/1	Running	0	17s

```

payal_kharat@LAPTOP-V1NV0QFT:kube$ curl http://localhost:8080/

```

```

<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>

```

```

<h1>http://info.cern.ch - home of the first website</h1>

```

```

<p>From here you can:</p>

```

```

<ul>

```

```

<li><a href="http://info.cern.ch/hypertext/WWW/TheProject.html">Browse the
first website</a></li>

```

```

<li><a href="http://line-

```

```

mode.cern.ch/www/hypertext/WWW/TheProject.html">Browse the first website using
the line-mode browser simulator</a></li>

```

```

<li><a href="http://home.web.cern.ch/topics/birth-web">Learn about the birth
of the web</a></li>

```

```

<li><a href="http://home.web.cern.ch/about">Learn about CERN, the physics
laboratory where the web was born</a></li>

```

```
</ul>
</body></html>
```

Question 6

Question-6:

Create 1 Public Docker Hub registry named cloudehix_hpa_yourname.

- Clone below repository on your system.
<https://github.com/vivekamin/kubernetes-hpa-example.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , build a docker image from the docker file and add meaningful tags and push to the docker hub repository.
- Once the image is pushed, go to k8s directory and update deployment.yaml file with image name from your repo. And then create it.
- Open service.yml and change the type to nodePort and apply the same.
- Open the HPA.yaml file, notice it and then apply the same.
- Open the browser, and access the webpage.
- Now it's time to test the HPA working with the below command.

kubectrl run -i --tty load-generator --rm --image=busybox --restart=Never --

```
/bin/sh -c "while sleep 0.01; do wget -q -O- http://NODE_PORT_SERVICE_NAME;
done"
```

- Check the HPA from kubectrl command and also check if the deployment is scaling up.
- Take the snap , prepare a well formatted doc and write your understanding.

=====

System / Terminal Output

```
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-6/.git/
-----
payal_kharat@LAPTOP-# git clone https://github.com/vivekamin/kubernetes-hpa-
example.git
Cloning into 'kubernetes-hpa-example'...
```

```
remote: Enumerating objects: 26, done.
remote: Total 26 (delta 0), reused 0 (delta 0), pack-reused 26 (from 1)
Receiving objects: 100% (26/26), done.
Resolving deltas: 100% (9/9), done.
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ cp -rf kubernetes-hpa-example/* .
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ ls
Dockerfile  README.md  k8s  kubernetes-hpa-example  package.json  src
-----

payal_kharat@LAPTOP-# podman build -t cloudethix-hpa:v1 .
STEP 1/6: FROM node:8.12.0-alpine
Resolved "node" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/node:8.12.0-alpine...
Getting image source signatures
Copying blob eeb7d76f44e7 done    |
Copying blob e35f88fcc259 done    |
Copying blob 4fe2ade4980c done    |
Copying config df48b68da0 done    |
Writing manifest to image destination
STEP 2/6: RUN mkdir -p /usr/src/app
--> 5adf40ad5b18
STEP 3/6: WORKDIR /usr/src/app
--> 57b6a4fbd200
STEP 4/6: COPY . /usr/src/app
--> 4fe5d69ed2ef
STEP 5/6: RUN npm install
npm notice created a lockfile as package-lock.json. You should commit this
file.
npm WARN node-app@1.0.0 No repository field.
-----

payal_kharat@LAPTOP-# podman tag cloudethix-hpa:v1
payalkharat/cloudethix_hpa_payal: hpa-v1
payal_kharat@LAPTOP-# podman push payalkharat/cloudethix_hpa_payal: hpa-v1
Getting image source signatures
Copying blob 93c832ae2e0b done    |
Copying blob 6fb9934f1d99 done    |
Copying blob 94a04f6c7594 done    |
Copying blob 4fe2ade4980c skipped: already exists
Copying blob eeb7d76f44e7 skipped: already exists
Copying blob e35f88fcc259 skipped: already exists
Copying config 2d73be4325 done    |
Writing manifest to image destination
-----

payal_kharat@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  replicas: 1
  selector:
    matchLabels:
      app: node-example
  template:
    metadata:
      labels:
        app: node-example
    spec:
```

```
containers:
- name: node-example
  image: payalkharat/cloudethix_hpa_payal:hpa-v1
  imagePullPolicy: Always
  ports:
  - containerPort: 3000
  resources:
    limits:
      cpu: "0.5"
    requests:
      cpu: "0.25"
```

```
payal_kharat@LAPTOP-# kubectl apply -f deployment.yml
```

```
deployment.apps/node-example created
```

```
payal_kharat@LAPTOP-# kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
node-example-65bb46b959-2pk4t	1/1	Running	0	2m41s

```
payal_kharat@LAPTOP-# cat service.yml
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: node-example
```

```
  labels:
```

```
    app: node-example
```

```
spec:
```

```
  selector:
```

```
    app: node-example
```

```
  ports:
```

```
  - port: 3000
```

```
    protocol: TCP
```

```
    nodePort: 30001
```

```
  type: NodePort
```

```
payal_kharat@LAPTOP-# kubectl apply -f service.yml
```

```
service/node-example createdpayal_kharat@LAPTOP-V1NV0QFT:k8s$ kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	118m
node-example	NodePort	10.96.107.252	<none>	3000:30001/TCP	27s

```
payal_kharat@LAPTOP-# cat hpa.yml
```

```
apiVersion: autoscaling/v1
```

```
kind: HorizontalPodAutoscaler
```

```
metadata:
```

```
  annotations:
```

```
  name: node-example
```

```
  namespace: default
```

```
spec:
```

```
  maxReplicas: 4
```

```
  minReplicas: 1
```

```
  scaleTargetRef:
```

```
    apiVersion: apps/v1
```

```
    kind: Deployment
```

```
    name: node-example
```

```
  targetCPUUtilizationPercentage: 1
```

```
payal_kharat@LAPTOP-# minikube addons enable metrics-server
```

💡 metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.

You can view the list of minikube maintainers at:

```
https://github.com/kubernetes/minikube/blob/master/OWNERS
  ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
🌟 The 'metrics-server' addon is enabled

-----

payal_kharat@LAPTOP-# kubectl apply -f hpa.yml
horizontalpodautoscaler.autoscaling/node-example created
payal_kharat@LAPTOP-# kubectl get hpa
NAME                REFERENCE                TARGETS          MINPODS   MAXPODS
REPLICAS   AGE
node-example    Deployment/node-example    cpu: 0%/50%      1          4          1
15s

-----

payal_kharat@LAPTOP-# curl http://localhost:8080
Hello World!!

-----

payal_kharat@LAPTOP-# kubectl run -i --tty load-generator --rm --image=busybox
--restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- http://node-
example:80; done"
All commands and output from this session will be recorded in container logs,
including credentials and sensitive information passed through the command
prompt.
payal_kharat@LAPTOP-# kubectl get hpa
NAME                REFERENCE                TARGETS          MINPODS   MAXPODS
REPLICAS   AGE
node-example    Deployment/node-example    cpu: 0%/50%      1          4          1
15s
payal_kharat@LAPTOP-# kubectl top pods
NAME                CPU(cores)   MEMORY(bytes)
node-example-86557bd844-rxnb6    1m           24Mi
```

Question 7

Question-7:

Create 1 Public Docker Hub registry named cloudethix_cronjob_yourname.

- Initialize a local repository & copy below code (three files) to your local repository in any of your working branch.
- Once code is copied, build the docker image from Dockerfile , add meaningful tags and then push the docker image to Docker hub registry.
- Now update the pythoncronjob.yml file to change the image name that you have just pushed to docker hub registry.
- Now create a cron job using pythoncronjob.yml file. Check with kubectl command if the cron job is created.
- Check the Job name which is created by cronjob from command line or lens.
- Then check the pod logs which are created by the job and capture the output.
- Prepare well formatted documents and write your understanding.

=====

System / Terminal Output

```
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
```

```
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignment-3/Que-7/.git/
```

```
-payal_kharat@LAPTOP-# podman build -t cloudethix-cronjob:1.0 .
STEP 1/7: FROM python:3.7-alpine
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:3.7-alpine...
Getting image source signatures
Copying blob 4819c95424fc done    |
Copying blob ea1518237b37 done    |
Copying blob 9875af95546d done    |
Copying blob 148762f75a1f done    |
Copying blob 96526aa774ef done    |
Copying config 1bac8ae77e done    |
Writing manifest to image destination
STEP 2/7: RUN addgroup -S appgroup && adduser -S appuser -G appgroup
--> 5951b5d6bc02
STEP 3/7: WORKDIR /app
--> bf7dfd535b22
STEP 4/7: COPY helloworld.py /app/helloworld.py
--> 7610f87d17b6
STEP 5/7: RUN chmod +x /app/helloworld.py
--> 467262f4ebfb
STEP 6/7: USER appuser
--> 8c5887a5bfe7
STEP 7/7: ENTRYPOINT ["python", "/app/helloworld.py"]
COMMIT cloudethix-cronjob:1.0
--> 4a23be180315
Successfully tagged localhost/cloudethix-cronjob:1.0
4a23be180315a733cd4b9363c761a666a3cce839df65009f864929e17585964e
```

```
payal_kharat@LAPTOP-# podman tag cloudethix-cronjob:1.0
payalkharat/cloudethix_cronjob_payal:cronjob-v1
payal_kharat@LAPTOP-V1NV0QFT:Que-7$ podman push
payalkharat/cloudethix_cronjob_payal:cronjob-v1
Getting image source signatures
Copying blob 8f516151e934 done    |
Copying blob 9875af95546d skipped: already exists
Copying blob ea1518237b37 skipped: already exists
Copying blob 96526aa774ef skipped: already exists
Copying blob 4819c95424fc skipped: already exists
Copying blob 148762f75a1f skipped: already exists
Copying blob 4ab828addb74 done    |
Copying blob 4ab828addb74 done    |
Copying config 4a23be1803 done    |
Writing manifest to image destination
payal_kharat@LAPTOP-# cat pythoncronjob.yml
```

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: python-helloworld
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: python-helloworld
              image: payalkharat/cloudethix_cronjob_payal:cronjob-v2
              command: ["/app/helloworld.py"]
          restartPolicy: OnFailure
payal_kharat@LAPTOP-# kubectl apply -f pythoncronjob.yml
cronjob.batch/python-helloworld created
payal_kharat@LAPTOP-# kubectl get cronjob
NAME                                SCHEDULE          TIMEZONE    SUSPEND    ACTIVE    LAST SCHEDULE
AGE
python-helloworld    */1 * * * *      <none>      False      0         27s
3m27s

-----

payal_kharat@LAPTOP-# kubectl get jobs
NAME                                STATUS    COMPLETIONS    DURATION    AGE
python-helloworld-29807272    Complete    1/1              9s          2m52s
python-helloworld-29807273    Complete    1/1              3s          112s
python-helloworld-29807274    Complete    1/1              3s          52s
payal_kharat@LAPTOP-# kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
python-helloworld-29807272-cmp74    0/1      Completed    0           2m57s
python-helloworld-29807273-zbknn    0/1      Completed    0           117s
python-helloworld-29807274-cgsgq    0/1      Completed    0           57s

-----

payal_kharat@LAPTOP-# kubectl logs python-helloworld-29807274-cgsgq
Welcome to the Cloudethix World
Today is
2026-09-04 07:39:00.852079
```